

Isolating the cause of quantum-kernel failure on malware family attribution

Short form

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Abstract

A fidelity-kernel quantum SVM (QSVM) falls to the chance level on the 15-class task of CIC-MalMem-2022 while reaching 0.42–0.52 macro-F1 on the equivalent EMBER 2018 task, with class imbalance and qubit count already excluded as causes. We complete the EMBER classical baseline with four models, validate the 1000-row subsample all quantum results rest on, measure class separability of the shared PCA projection, and measure kernel concentration and alignment against an RBF control built on the identical feature matrix. Random forest beats SVM on every EMBER binary configuration, so the recorded classical advantage was understated by 0.011 to 0.027 macro-F1. All 84 comparable pairs favour the classical model at $p < 0.05$. The RBF control produces the same off-diagonal Gram spread on both corpora (0.2212 against 0.2184), whereas the fidelity kernel matches RBF on EMBER ($0.92\times$) and collapses to $0.40\times$ on CIC. The collapse is a property of the kernel, not of the corpus.

1 Introduction

Three questions were open. The EMBER classical arm used a support vector machine only, so every recorded margin there compared against one baseline rather than the strongest available. The QSVM’s collapse on CIC 15-class was unexplained. And every claim of a classical advantage rested on differences in mean macro-F1 with no test that a gap exceeded fold-to-fold noise. A fourth issue was that all quantum results are computed on a 1000-row subsample whose fidelity to its population had been assumed rather than measured.

2 Materials and methods

Data. *CIC-MalMem-2022*: memory dumps, 55 forensic features; 58 596 balanced rows for the binary task and 29 298 malware-only rows over 15 families ($1.7\times$ imbalanced) for the 15-class task. *EMBER 2018*: static PE features, 2381 dimensions, on an 18 014-row extract of the published test set (88.9% malware); its 15-class task keeps the 15 largest families and downsamples each to the smallest, giving 14 325 rows at 955 per family.

Protocol. Both frameworks consume one pipeline: variance filter, correlation filter at 0.95, standardisation, PCA to n_c components, fit on training-fold rows only. n_c abbreviates $n_{\text{components}}$ and sets the classical feature count and the quantum qubit budget simultaneously. The fidelity kernel costs one circuit evaluation per training pair, so quantum runs use a stratified 1000-row subsample; that subsample and its five outer folds are persisted by the quantum runner and replayed by the classical one, giving 800 training and 200 held-out rows per fold.

Metric. Held-out macro-F1 averaged over five folds. Accuracy is not reported: on EMBER binary at $n_c = 1$ the QSVM reaches 0.890 accuracy in three of five folds while the benign class’s F1 is exactly 0.000, because 88.9% of rows are malware and the model labels everything malware.

Measurements introduced here. Four classical models at $n_c \in \{1, 3, 6\}$ on both EMBER tasks; per-feature Kolmogorov–Smirnov tests of the subsample against its population; Fisher ratio and silhouette of the shared projection; off-diagonal Gram spread and kernel-target alignment for the fidelity kernel and for an RBF control on the identical matrix; paired t -test and Wilcoxon over per-fold macro-F1.

Run selection. Fold records are grouped by cross-validation sweep, not by configuration. The same configuration was swept more than once, so a parameter-based grouping returns 10, 15 or 20 records for a five-fold sweep, and for the QSVM the encoding is a per-fold tuner outcome rather than a configuration key. Only sweeps completing with exactly five folds are retained: 123 of 125.

Table 1: EMBER 2018 held-out macro-F1, mean over five outer folds. Best classical model in bold.

	Random forest	XGBoost	LightGBM	SVM	QSVM angle	QSVM IQP
<i>Binary</i>						
$n_c = 1$	0.5582	0.5451	0.5093	0.5314	0.4532	0.4587
$n_c = 3$	0.6575	0.6465	0.6434	0.6306	0.4924	0.5084
$n_c = 6$	0.6835	0.6680	0.6593	0.6724	0.5138	0.5137
<i>15-class</i>						
$n_c = 1$	0.5562	0.5313	0.5410	0.5294	0.1688	0.1541
$n_c = 3$	0.7194	0.7089	0.7035	0.7389	0.3908	0.4797
$n_c = 6$	0.7903	0.7659	0.7709	0.7930	0.4232	0.5195

3 Results

3.1 The classical baseline was incomplete

Random forest beats SVM on every binary configuration, widening the classical advantage from +0.0727 to +0.0995 at $n_c = 1$, from +0.1222 to +0.1491 at 3, and from +0.1587 to +0.1697 at 6. On 15-class SVM is the strongest baseline at $n_c \in \{3, 6\}$, so those margins stand. Four of six configurations move, all in the same direction.

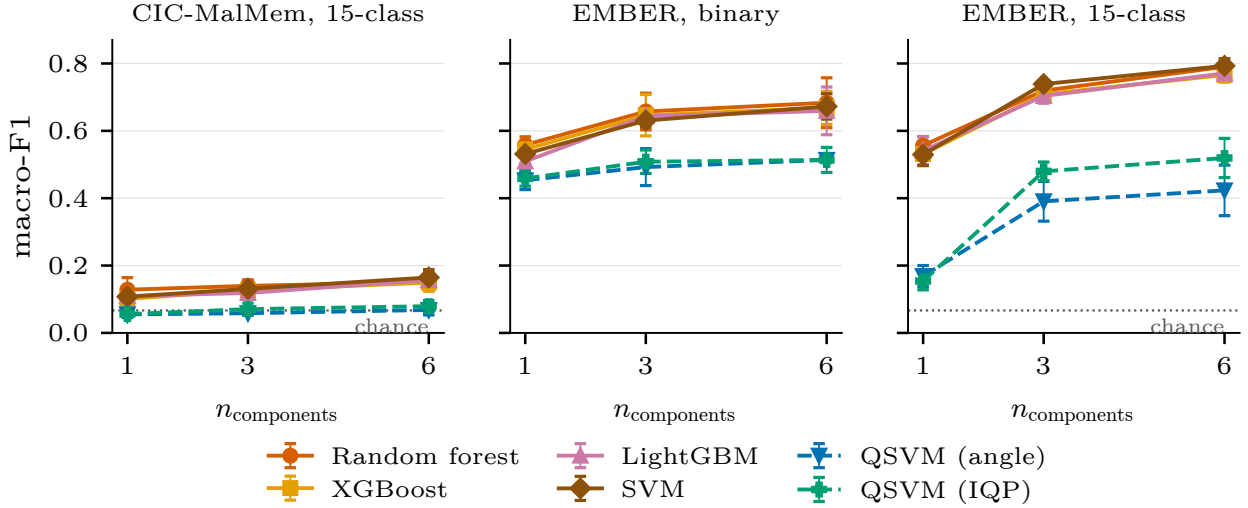


Figure 1: Macro-F1 against the shared feature and qubit budget; solid lines classical, dashed the QSVM encodings, dotted the 15-class chance level. On EMBER binary the QSVM gains 0.055 macro-F1 from one to six components where the random forest gains 0.125, so the gap widens with capacity. CIC binary is not shown, for the reason in §4.1.

3.2 The subsample preserves its population

A rejection count is interpretable only against its null expectation, since a faithful subsample still rejects about $\alpha \cdot n_{\text{features}}$ features by chance. CIC rejects **0** of 55 where ≈ 2.75 were expected, EMBER **1** of 2381 where ≈ 119 were. Maximum class-proportion drift is below 0.001 on both.

3.3 Feature geometry accounts for the corpus, not the kernel

EMBER’s families are $\approx 8\times$ more separable by Fisher ratio and up to $50\times$ on the worst-case family. CIC’s silhouette is negative at every n_c , so the average point lies closer to another family’s centroid than its own. CIC’s PCA nonetheless explains more variance (33–81% against 8–30%), placing its variance in directions that do not discriminate family.

This accounts for the poor absolute performance of every method on CIC 15-class. It does not account for a QSVM-specific deficit, since a classical SVM extracts 0.108–0.165 macro-F1 from these same features where the QSVM extracts 0.056–0.079.

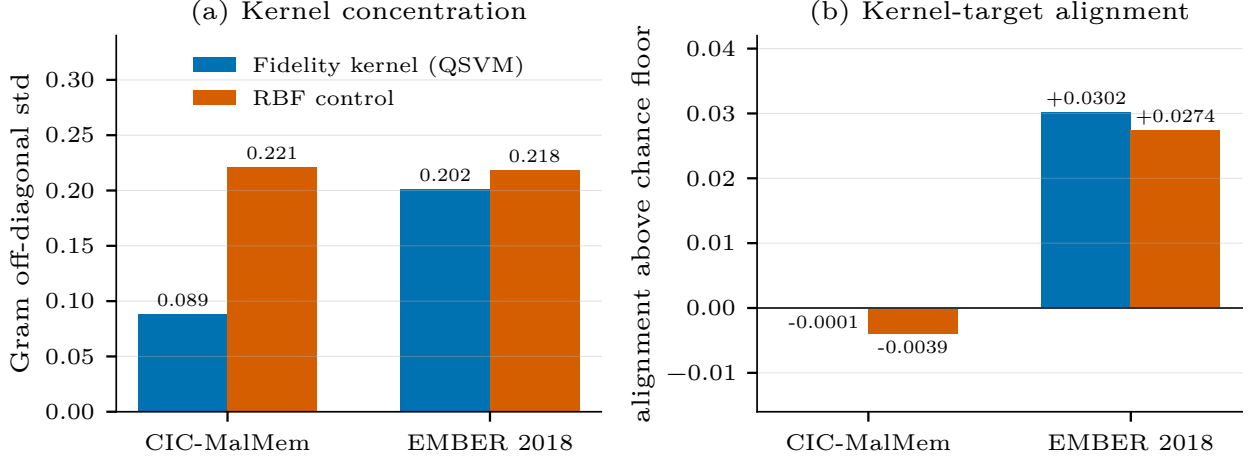


Figure 2: First outer training fold, $n_c = 6$, IQP encoding, 800 rows, with an RBF control on the identical feature matrix. (a) Off-diagonal Gram spread. (b) Alignment relative to its floor, a constant kernel scoring $1/\sqrt{15} \approx 0.258$ on 15 balanced classes.

3.4 Kernel concentration accounts for the kernel

The RBF control produces near-identical spread on both corpora, so CIC’s features do not cause a classical kernel to concentrate. The fidelity kernel tracks RBF on EMBER at $0.92\times$ and collapses to $0.40\times$ on CIC, compressing every off-diagonal similarity into a band $2.5\times$ narrower from the same inputs.

On alignment the CIC fidelity kernel lands 0.000069 *below* the constant-kernel floor, indistinguishable from a matrix of ones, and the RBF control is also below it; on EMBER both clear it by ≈ 0.03 . Alignment separates the corpora while concentration separates the kernels. Across the full sweep EMBER’s Gram is $1.9\text{--}2.6\times$ more spread than CIC’s at every comparable cell.

3.5 The differences are statistically significant

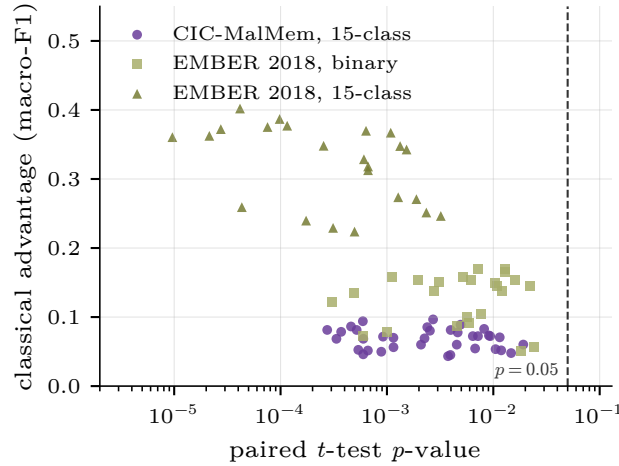


Figure 3: All 84 comparable quantum–classical pairs. Every point lies left of $p = 0.05$ and above zero.

All 84 comparable pairs favour the classical model at $p < 0.05$; the largest p -value is 0.0241. Wilcoxon returns exactly 0.0625 for all 84, which at five paired samples is the smallest attainable two-sided value, so the test cannot reach 0.05 at this fold count regardless of effect size. Among classical models on shared predictions, McNemar separates random forest from XGBoost on EMBER 15-class at $p = 0.0022$, while on CIC 15-class no classical pair separates at all.

4 Discussion

Feature geometry and kernel geometry give different answers, and that difference is the diagnosis. Geometry explains why every method performs poorly on CIC 15-class but cannot explain a QSVM-specific deficit,

since the classical SVM recovers two to three times the QSVM’s macro-F1 from the same projection. Kernel concentration does explain it: the fidelity kernel loses 60% of the RBF control’s off-diagonal spread on inputs where the control loses none. A Gram approaching a scaled identity presents the SVM with a geometry in which all pairs are equally similar, which is what a macro-F1 pinned to chance looks like internally. The result is a statement about the method rather than the corpus.

4.1 Limitations

- **Bandwidth is untested on CIC.** Concentration is the identified mechanism and bandwidth is the parameter that acts on it, but no sweep was run. §3.4 identifies a mechanism, not a remedy.
- **Alignment is a weak predictor.** It reports that neither kernel carries label information on CIC, yet a classical SVM reaches 0.16 there. Alignment is a global unweighted average over all pairs and an SVM requires no such property, so it ranks corpus difficulty usefully and predicts attainable performance poorly.
- **Scope of the kernel measurement.** The alignment and RBF-control figures come from one training fold, one capacity and one encoding, with a single untuned γ for the control.
- **One configuration is not separable.** Every CIC binary run, classical and quantum alike, predates sweep-level tracking and carries no recoverable boundary; 45 fold records per model resolve into two sweeps per n_c with no identifier separating them. No grouping was reconstructed. The limitation is symmetric across frameworks.
- **No sample-level test between frameworks.** McNemar needs per-sample predictions from both models; QSVM predictions were not persisted for these sweeps, so cross-framework evidence rests on five paired fold-level scores.
- **Fold count.** Five folds support the paired t -test and are too few for Wilcoxon. Six would place 0.03125 within reach.
- **Simulator only.** No hardware execution and no noise model.

5 Conclusion

Completing the classical baseline widened the recorded classical advantage on EMBER binary by 0.011 to 0.027 macro-F1, and paired testing establishes that all 84 comparable differences reach $p < 0.05$. The subsample every quantum result depends on is faithful to its population. The QSVM’s collapse on CIC 15-class is attributable to kernel concentration rather than to feature difficulty: an RBF kernel on the identical matrix retains its spread while the fidelity kernel falls to $0.40\times$ of it, which locates the failure in the kernel and names the parameter that acts on it.